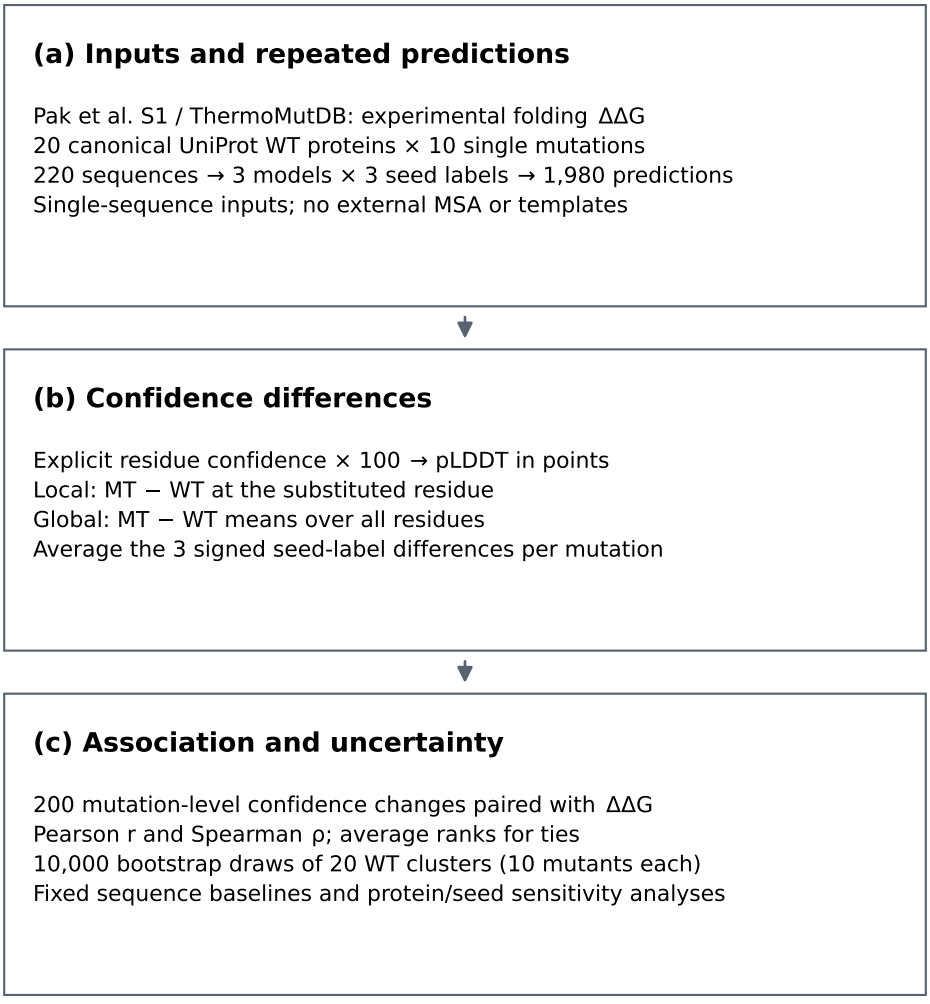
(a) Inputs and repeated predictions
Pak et al. S1 / ThermoMutDB: experimental folding ΔΔG
20 canonical UniProt WT proteins × 10 single mutations
220 sequences → 3 models × 3 seed labels → 1,980 predictions
Single-sequence inputs; no external MSA or templates
(b) Confidence differences
Explicit residue confidence × 100 → pLDDT in points
Local: MT − WT at the substituted residue
Global: MT − WT means over all residues
Average the 3 signed seed-label differences per mutation
(c) Association and uncertainty
200 mutation-level confidence changes paired with ΔΔG
Pearson r and Spearman ρ; average ranks for ties
10,000 bootstrap draws of 20 WT clusters (10 mutants each)
Fixed sequence baselines and protein/seed sensitivity analyses